Class 32: E Vs k, Brillouin Zones and the Origin of Bands

In this class we will plot the reciprocal lattice information as well as the wave vector information on the same plot and examine the interaction between them in a pictorial manner.

We have noted that while the $E\ Vs\ k$ relationship is the same for free electrons, nearly free electrons, as well as the bound electrons, it is a continuous curve only for free electrons. For nearly free electrons as well as bound electrons, confinement of the electron results in only specific values of k to be permitted, and therefore only the corresponding values of energy to be permitted. The resulting plot has discrete points that are laid out in the form of the parabola consistent with

$$E = \frac{\hbar^2 k^2}{2m}$$

We wish to plot the periodicity of the lattice as well as the wave vectors on the same plot. The reciprocal space differs from k space only in the scaling factor 2π. Therefore reciprocal lattice information is multiplied by 2π to enable it to be plotted in k space. Therefore real lattice vector 'a' which is plotted in reciprocal space as $\frac{1}{a}$, is now plotted in k space as $\frac{2\pi}{a}$. We note that a, which of the order of inter-atomic spacing, is of the order of 10^{-10} m, while L, the extent of confinement of the nearly free electrons, is of the order of meters. Therefore the reciprocal lattice points, plotted in k space as $\frac{2n\pi}{a}$, where n is an integer, can contain 10^{10} allowed wave vectors of nearly free electrons which are of the form $\frac{n\pi}{L}$ between two adjacent reciprocal lattice points. This is the reason that allowed wave vector plots drawn on the scale of the order of $\frac{2n\pi}{a}$, will look continuous even though they contain discrete points spaced $\frac{\pi}{L}$ apart.

We have noted that diffraction occurs when one end of a wave vector touches a Bragg plane. In the context of nearly free electrons, diffraction causes the $E\ Vs\ k$ relationship to distort in the vicinity of the Bragg planes (or Brillouin zone boundaries). This distortion causes some of the energy levels to become forbidden to the nearly free electrons – which results in the presence of band gaps in the material.

In this class we will present this interaction between wave vectors and the Brillouin zones pictorially and not focus on the exact values of the band gaps that result. We will look at examples of one dimensional, two dimensional, and three dimensional lattices and the interaction of wave vectors with the Brillouin zones corresponding to these lattices in k space. In the next class we will examine the formation of band gaps in a mathematical manner.

Since a one dimensional real lattice of spacing 'a' is plotted in k space as $\frac{2\pi}{a}$, the Brillouin zone boundaries occur in intervals of $\frac{\pi}{a}$, i.e. at $\frac{\pi}{a}$, $\frac{2\pi}{a}$, $\frac{3\pi}{a}$ etc. The Figure 32.1 below shows the interaction of the $E\ Vs\ k$ relationship of nearly free electrons with the Brillouin zone boundaries of a one dimensional lattice of spacing a

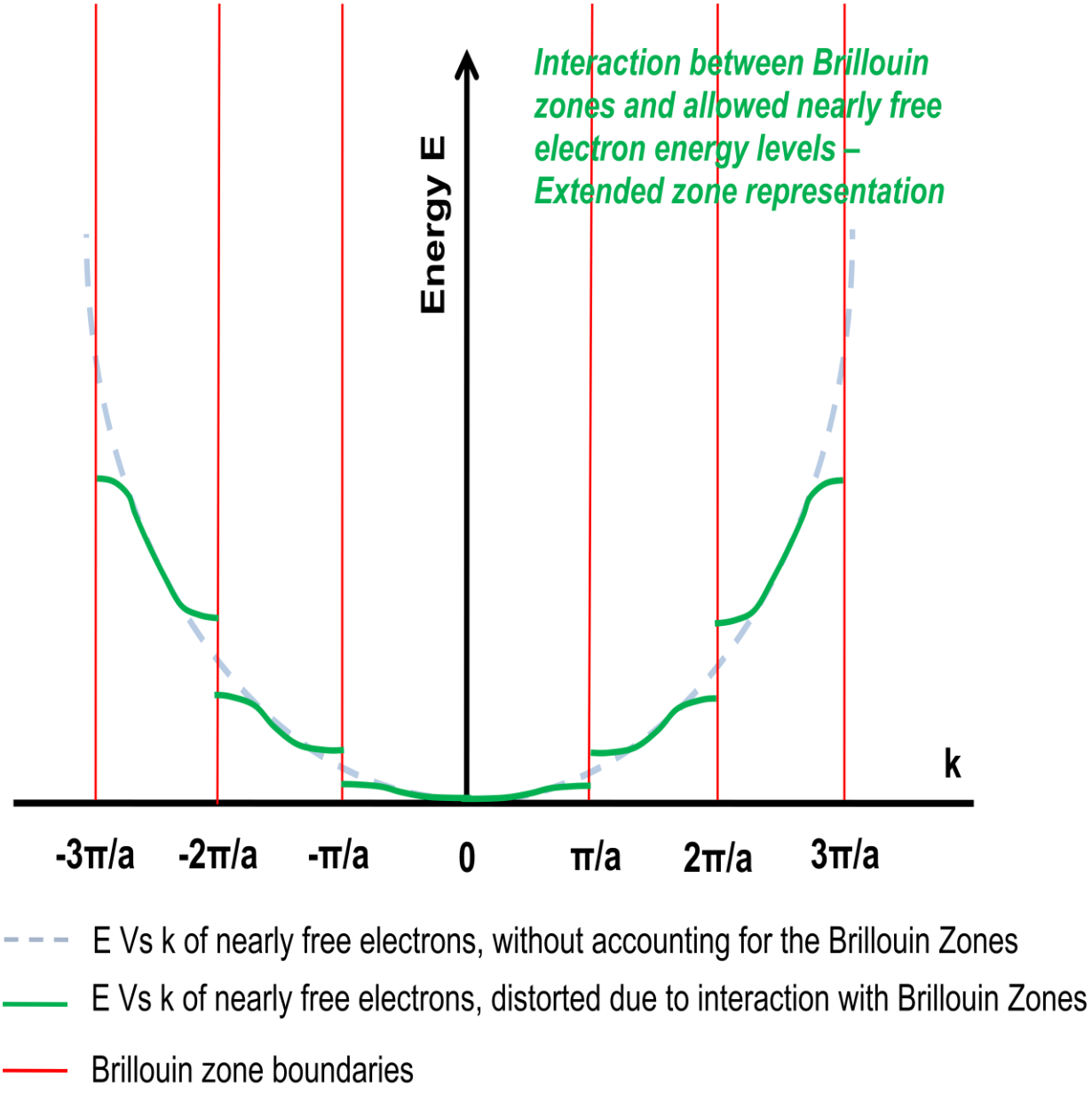


Figure 32.1: Interaction of the $E\ Vs\ k$ relationship of nearly free electrons with the Brillouin zone boundaries of a one dimensional lattice of spacing a

The distortion of the $E\ Vs\ k$ relationship at the Brillouin zone boundaries and the resulting energy gaps, can be described as follows: the travelling waves of nearly free electrons undergo diffraction at the Brillouin boundaries and result in standing waves – as a result energy gaps appear.

The above form of representing the interaction between the allowed energy values and the Brillouin zones of the material is called the 'Extended zone' representation, since the information is presented spread across several lattice points, but has a single origin.

The band gaps described here, and the remaining allowed energy levels, are in concept the same as the band gaps and the allowed bands, we are familiar with in the band structure of materials. However, in high school texts, bands of allowed energy levels are represented as boxes, separated by gaps which represent the band gaps, and such a diagram is called a 'Flat band diagram'. How does the flat band diagram relate to Figure 32.1 above? This relationship is shown in Figure 32.2 below.

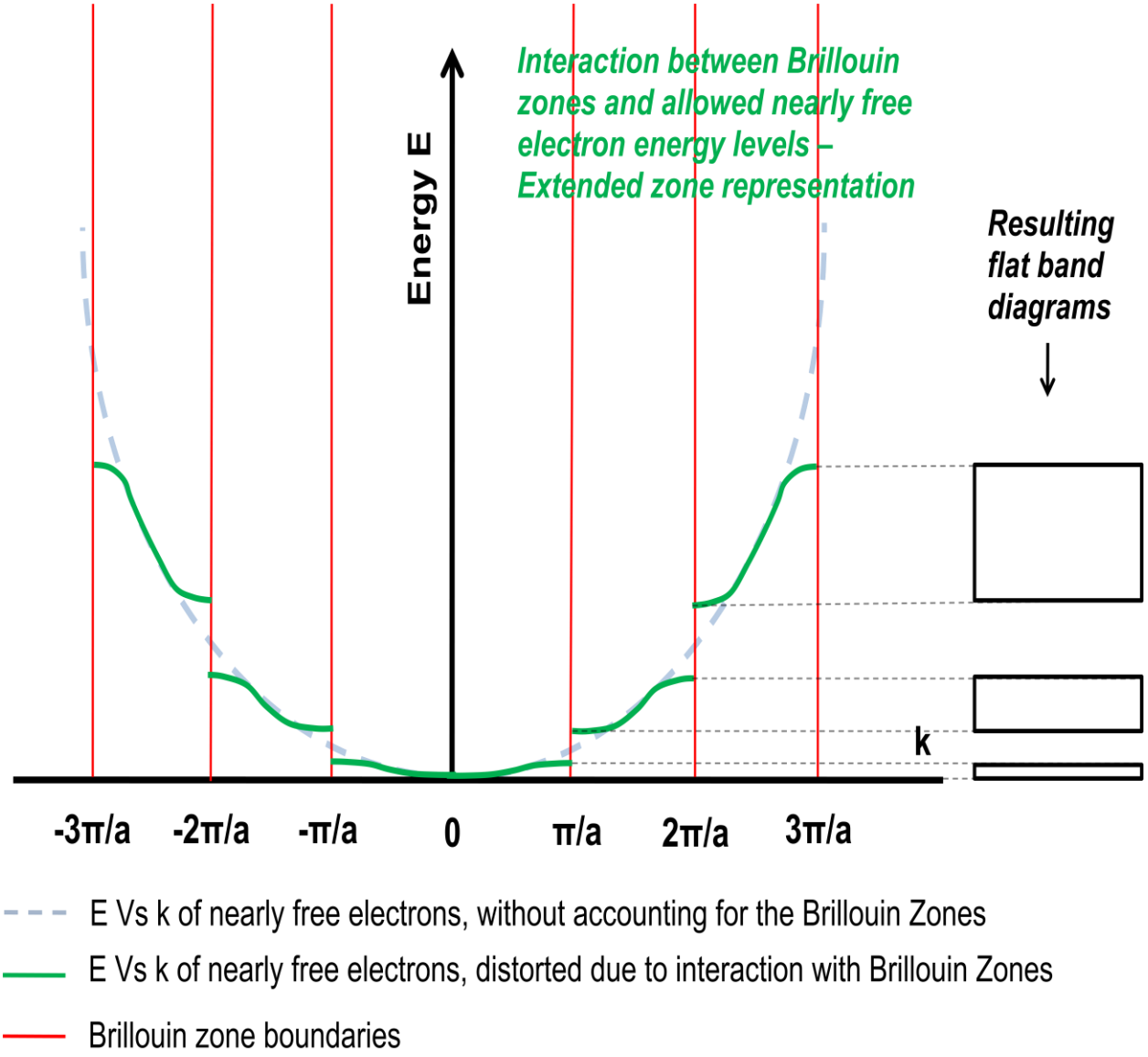


Figure 32.2: Relationship between the extended zone representation and the flat band structure.

The flat band diagram is sufficient to explain specific material phenomena, but is insufficient to explain many others. The other representations we are presenting in this class, including the extended zone representation and some more that will follow, are much more capable of explaining a variety of material phenomena when compared to the flat band diagram.

We have also noted that at 0 Kelvin the Fermi energy represents the highest occupied energy level in the system, and interaction of electrons with the outside world begins here. Where does this fit into our representations of the allowed energy levels, and how does that relate to the flat band diagram? Figure 32.3 below shows an example of this.

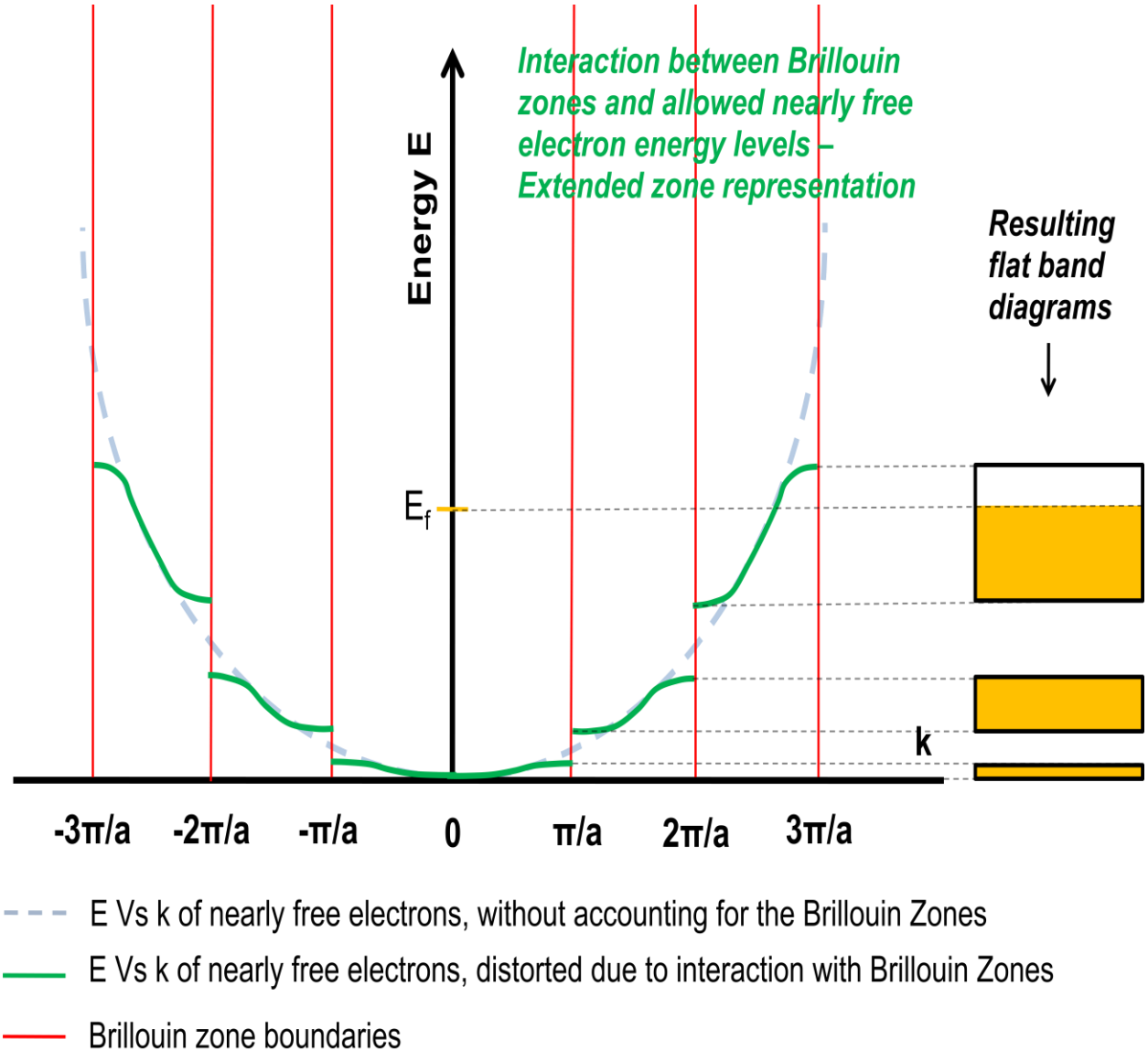


Figure 32.3: The Fermi energy in the extended zone scheme and in the flat band diagram.

While the extended zone scheme plots the essential details, it is important to note that the choice of origin of k space is arbitrary. Due to symmetry, each lattice point is the same as every other lattice point. Therefore the $E\ Vs\ k$ diagram can be repeated at each lattice point, to give a more complete picture of the situation in the material. This type of a diagram is called a 'Repeated zone' representation, and is shown in Figure 32.4 below.

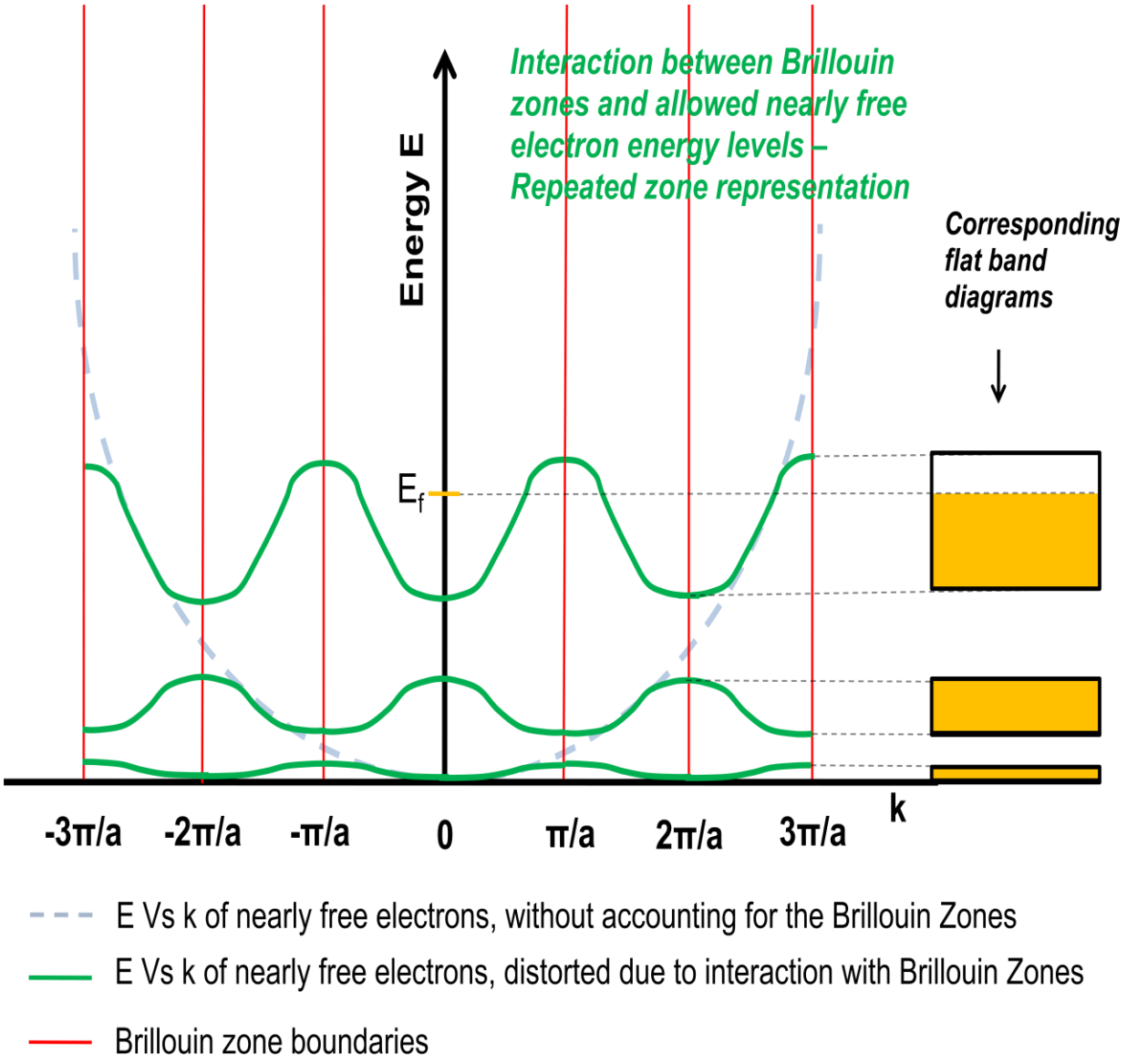


Figure 32.4: The repeated zone representation

Looking at the repeated zone representation, it can be concluded that the region in the first Brillouin zone, between $-\frac{\pi}{a}$ and $\frac{\pi}{a}$, contains all of the information in a compact manner. Therefore it is often considered sufficient to show all of the allowed wave vectors and energy levels within the first Brillouin zone itself. Such a representation is referred to as the 'Reduced zone' representation, and is as shown in Figure 32.5 below.

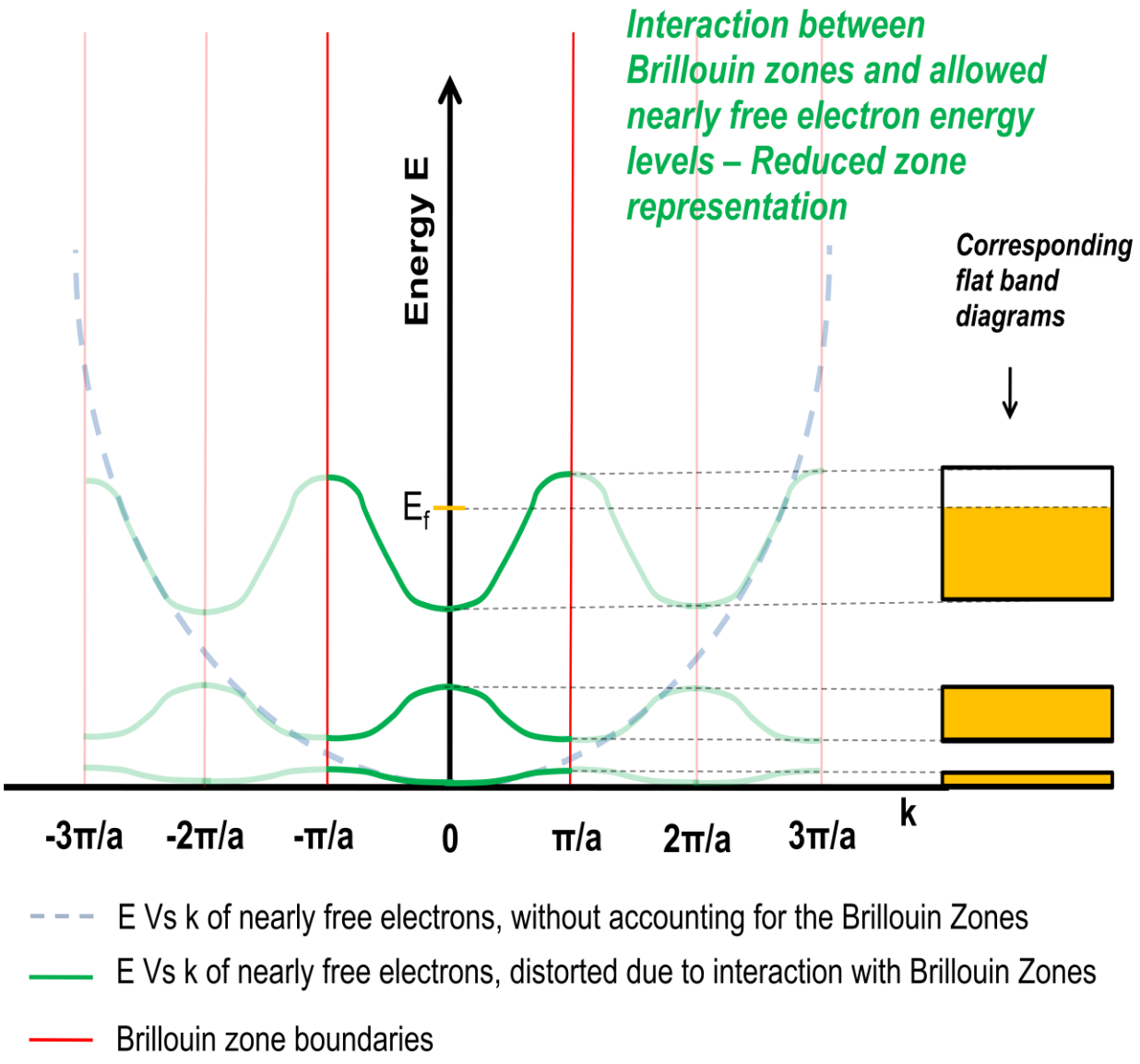


Figure 32.5: The reduced zone representation

In all of these diagrams, it is the location of the Fermi energy, E_f, which decides whether the material is metallic, semiconducting, or insulating. In the diagrams drawn here, E_f is in the middle of a band, therefore the material being depicted is metallic.

The above diagrams were for a one dimensional lattice. Let us consider a two dimensional square lattice and schematically consider its interaction with the wave vectors of nearly free electrons. In two dimensional k space, electrons of the same energy are represented by a circle. The Fermi energy corresponds to the largest such circle, and we are therefore interested its interaction with the two dimensional Brillouin zone boundaries. Figure 32.6 below shows the first two Brillouin zones of a two dimensional lattice and consider four possible values of N_f, resulting in four different values of E_f, and hence circles of corresponding diameters. As long as the circle is far

from the Brillouin zone boundary, it does not interact with the boundary, and remains undistorted. When the circle gets close to the boundary, diffraction effects cause it to distort. When the circle is slightly larger than the Brilloun zone, distorted sections of the circle appear inside the first Brillouin zone as well as in the second Brillouin zone.

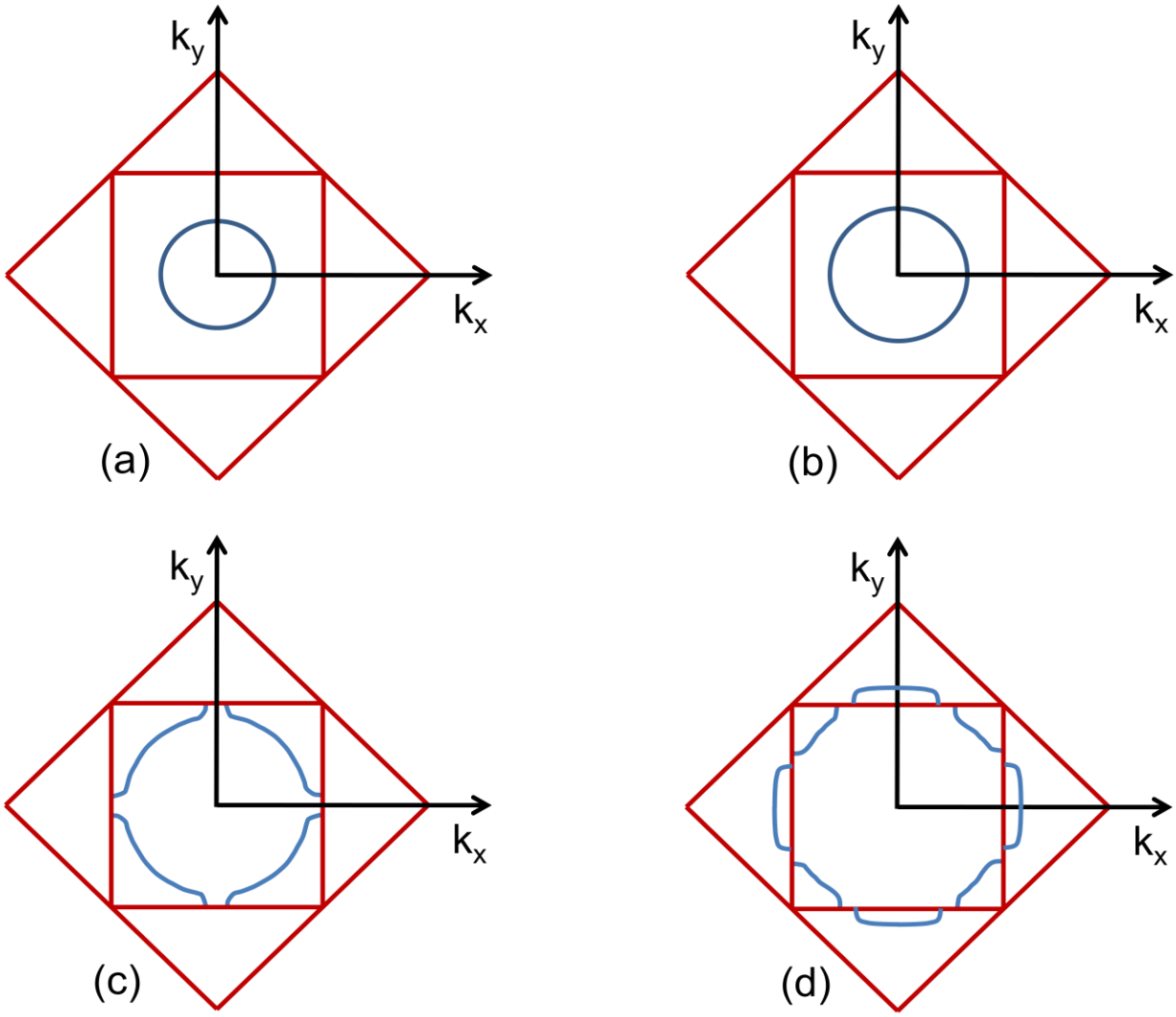


Figure 32.6: Interaction between allowed wave vectors of nearly free electrons and Brillouin zones of a two dimensional square lattice. The figures (a), (b), (c), and (d) differ in the value of N_f being depicted. N_f in (a) < N_f in (b) < N_f in (c) < N_f in (d). The allowed wave vectors are contained within the first Brillouin zone in figures (a), (b), and (c), In figure (d) the allowed wave vectors appear in the first as well as the second Brillouin zones.

In two dimensions, as well as in three dimensions, it is possible to plot in the reduced zone representation, in just the manner it was accomplished in the one dimensional case. The figures get complicated but the concept is the same. By moving higher Brillouin zones by valid lattice vectors in k space, the information corresponding to higher Brillouin zones can be represented in

the first Brillouin zone itself. Figure 32.7 below shows this for the two dimensional square lattice.

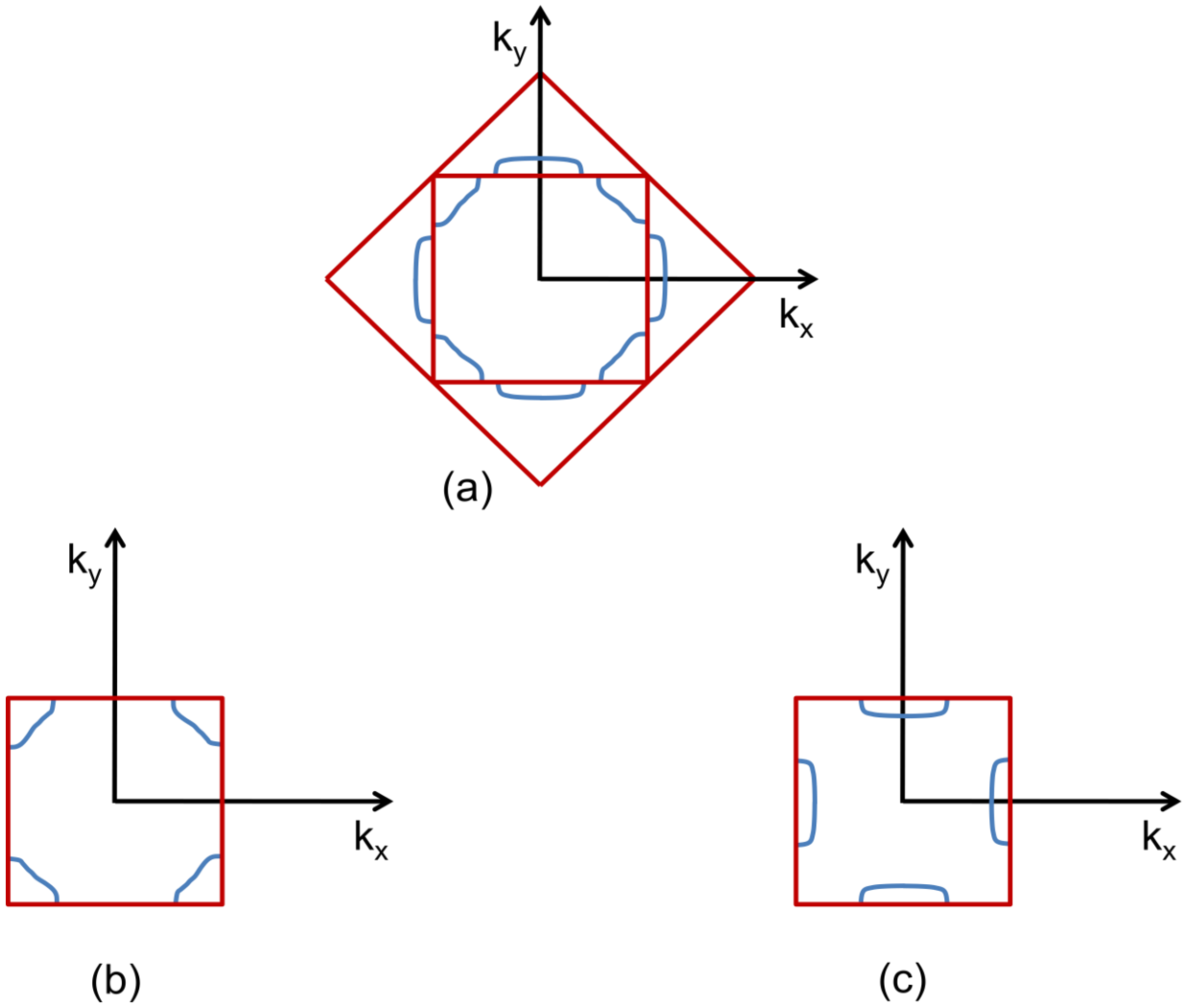
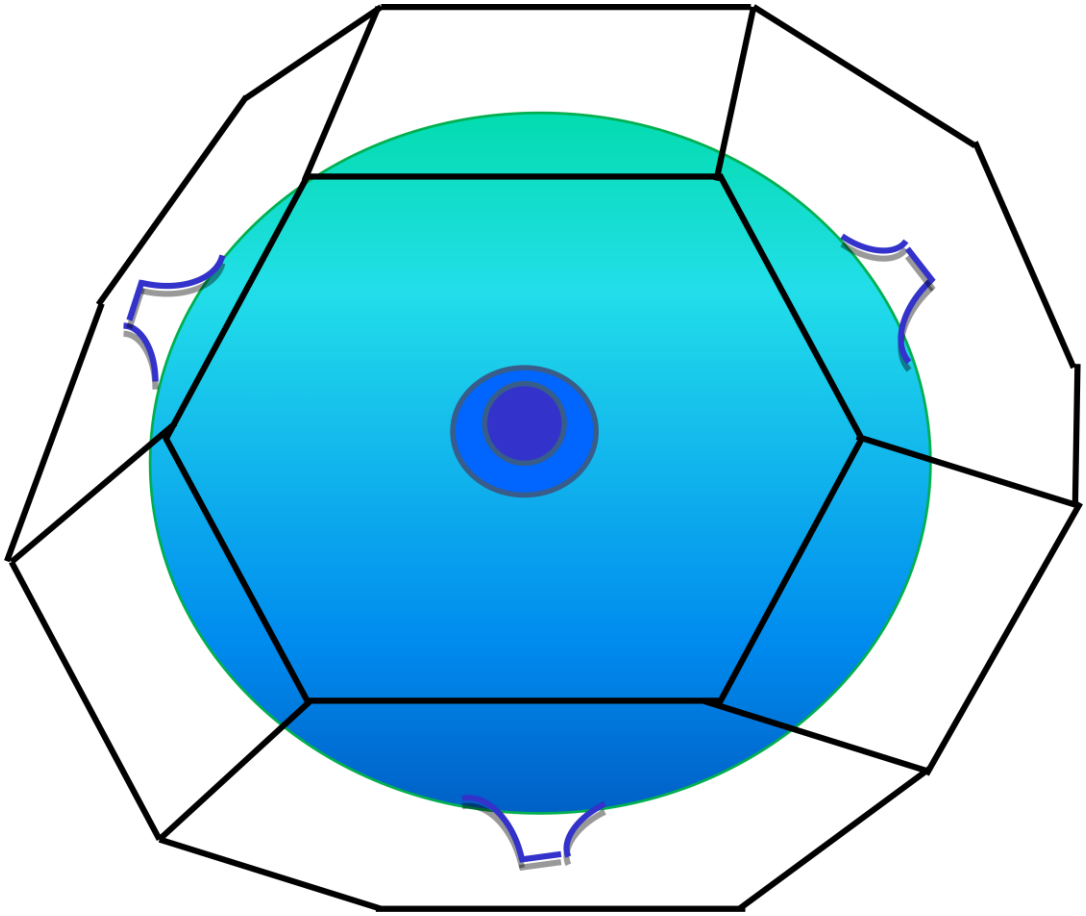


Figure 32.7: (a) Extended zone representation, (b) and (c) Reduced zone representation

Finally, we can also look at an example of a material that has an FCC real lattice, and therefore a Brillouin zone that is the Wigner Seitz cell of a BCC lattice. Spheres will indicate states of the same energy and based on the value of N_f, the largest possible sphere will indicate the E_f, and is the Fermi sphere or Fermi surface. Figure 32.8 below schematically shows a case where the sphere corresponding to the Fermi energy just touches the Brillouin zone in specific directions. If N_f were lower, the sphere would not touch the Brillouin zone at any location, and if N_f were higher, the sphere would distort and extend into the second zone in specific directions.



Animation of the above is shown in the next page

Figure 32.8: The interaction between the Brillouin zone of an FCC real lattice, with the Fermi sphere

In this class we have seen the interaction between the Brillouin zones and the Fermi surface pictorially, and seen how this results in the Band structure of the materials. In the next class we will look at this interaction in an analytical manner.

Animation of figure 32.8

